

SALITHA MARASINGHE

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SUMMARY

Final-year Data Science undergraduate with hands-on industry experience as a Data Engineer intern, bringing a track record of building and deploying data engineering pipelines, machine learning models, statistical analytics pipelines, and AI-integrated systems across domains and business contexts. Combines strong theoretical foundations in statistical modelling and applied ML with practical experience delivering end-to-end data solutions in Agile, cross-functional environments.

SKILLS

Languages: Python | SQL

ML & AI: Scikit-Learn | NumPy | Pandas | Matplotlib | Time Series Forecasting | Ensemble Methods | Supervised & Unsupervised Learning | SHAP | LLM Integration

Statistical & Analytical: Feature Engineering | Statistical Modelling | Predictive Analytics | Model Evaluation & Tuning

Data Engineering, Cloud & BI: Microsoft Azure | Microsoft Fabric | Azure Data Factory | Azure Data Lake Gen2 | Databricks | ETL/ELT Pipeline Design | Medallion Architecture | Power BI | Microsoft Excel |

Databases: SQL Server | MySQL | Firebase | MongoDB

Software & DevOps: Fast API | React | GitHub | BitBucket | Railway | Firebase

Methodologies & Practices: Cross-Functional Agile Collaboration | End-to-End System Ownership | Technical Team Leadership | Proactive Problem Solving

EXPERIENCE

Data Engineer Intern • TRACE – Sri Lanka | Remote • Aug 2025 – Feb 2026

- Built an automated Python ETL pipeline (Customs Data Transformer) to ingest, validate, and transform Excel manifests into standards-compliant XML, eliminating a manual enterprise data submission process.
- Designed ER schemas and implemented the backend for a multi-source B2B pricing platform (Wasi Lanka) and conducted market research on feasibility to support strategic decision making.
- Designed an LLM-powered AI data agent (Tourism AI Agent) with MCP-agent design and consent-based behavioral data collection pipelines.
- Adhered to DevOps principles by enforcing Git/BitBucket version control, rigorous code review standards, and continuous integration workflows across codebases.
- Acted as team lead within a cross-functional Agile delivery environment, managing full-lifecycle sprint deliverables from development and QA through to deployment, while translating complex data pipeline architectures into clear outputs for non-technical business stakeholders.

PROJECTS

Sri Lanka Tourism Intelligence Platform | Research Project

Tech Stack: Python, Fast API, React, Firebase, Railway

- Engineered a multi-source data ingestion pipeline integrating macroeconomic indicators, behavioral signals (Google Trends), and weather covariates to support a suite of predictive analytics models.
- Trained and evaluated 11 models across Time Series, ML, and Deep Learning paradigms for national arrival forecasting; selecting best-performing base learners, optimizing each via Bayesian Optimization, and searching ensemble weights using a Genetic Algorithm.
- Extended the forecasting suite with revenue intelligence and source market segmentation models, and integrated SHAP-based explainability to surface actionable feature-level drivers for non-technical decision-makers.
- Built three automated LLM-based geopolitical intelligence agents with weekly refresh, independently analyzing global event impacts on arrival forecasts, revenue risk, and source market intelligence to support strategic decision-making.

ISP Customer Analytics & Churn Prediction Platform

Tech Stack: Azure Data Factory, Databricks, PySpark, SQL, Scikit-learn, Power BI, Medallion architecture

- Engineered an end-to-end cloud Lakehouse pipeline on Azure (ADF + Databricks + ADLS Gen2) unifying multi-domain customer, network, support, and billing data through Bronze/Silver/Gold Medallion architecture.
- Designed a Star schema dimensional model in the Gold layer, exposing curated Delta tables via Databricks SQL Warehouse to support real-time descriptive and predictive analytics reporting.
- Applied supervised ML techniques (Logistic Regression, Gradient Boosting, Random Forest) to build a customer churn prediction model from engineered analytical features from complex network, support, and billing signals to identify at-risk customers.
- Performed statistical analysis across customer segments, SLA performance, and billing behavior to surface actionable trends and measure operational performance and extended insights to an ML-driven churn probability dashboard that surfaces individual customer risk scores via live Power BI.

E-Commerce Customer Churn Prediction System

Tech Stack: Python, Scikit-learn, Flask

- Contributed to Engineering RFM-based behavioral features (Recency, Frequency, Monetary, engagement level, return rate) from 250,000 e-commerce transaction records to support churn classification.
- Contributed to Benchmarking 14 classification algorithms across the full ML spectrum, selecting and tuning a Decision Tree via RandomizedSearchCV; achieving 93.2% accuracy and 0.83 F1-score on a class-imbalanced held-out test set.

E-Commerce Product Recommendation System

Tech Stack: Python, Scikit-learn, Flask, MySQL

- Contributed in implementing two complementary unsupervised recommendation strategies; content-based filtering (TF-IDF + Euclidean distance on product tags) and user-based collaborative filtering (cosine similarity on a user-item rating matrix) to deliver personalized product discovery.

CropIQ Smart Soil Health Prediction System

Tech Stack: Python, Firebase Realtime Database, Scikit-learn, Fast API, React

- Engineered a real-time IoT data ingestion pipeline processing continuous environmental sensor telemetry (NPK, temperature, humidity, soil moisture) into Firebase Realtime Database, and built a waterlogging risk prediction model
- Delivered predictive insights and real-time sensor analytics through a full-stack Fast API + React dashboard integrated with a Gemini-powered AI advisory chatbot.

EDUCATION

BSc (Hons) Information Technology – Specializing in Data Science (Undergraduate)

Sri Lanka Institute of Information Technology (SLIIT) | 2022 – Present

- Relevant Coursework: Database Warehousing and Business Intelligence (DWBI), Machine Learning and Optimization Methods (MLOM), Theory and Practices in Statistical Modelling (TPSM), Introduction to Internet of Things and Big Data Analytics (IoTBD)

REFERENCES

Lahiru Wijesekara

Senior Solutions Architect | Expert in Enterprise Applications & Integration | Software Consultant | Academic & Industry Researcher

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